Sept 23rd 2026

Theoretical Particle Physics Seminar
University of Melbourne
Melbourne, AUS

The Preference for Evolving Dark Energy through a minimal Density-Level Parameterization

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Wisconsin IceCube Particle Astrophysics Center (WIPAC),
University of Wisconsin, Madison

Based on
2603.25735 with Richard Stiskalek
2509.13318 with Michael Toomey, Evan McDonough and Katherine Freese



The puzzling accelerated expansion of the Universe

Gravity is attractive so the expansion should be slowing down.



JWST Deep Field

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- Instead we observe **acceleration**



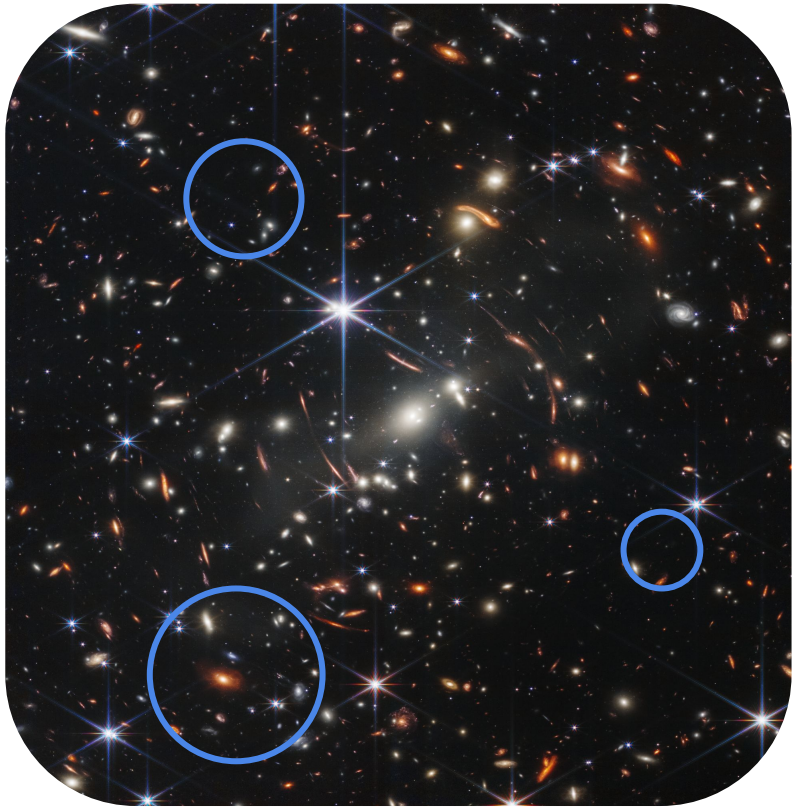
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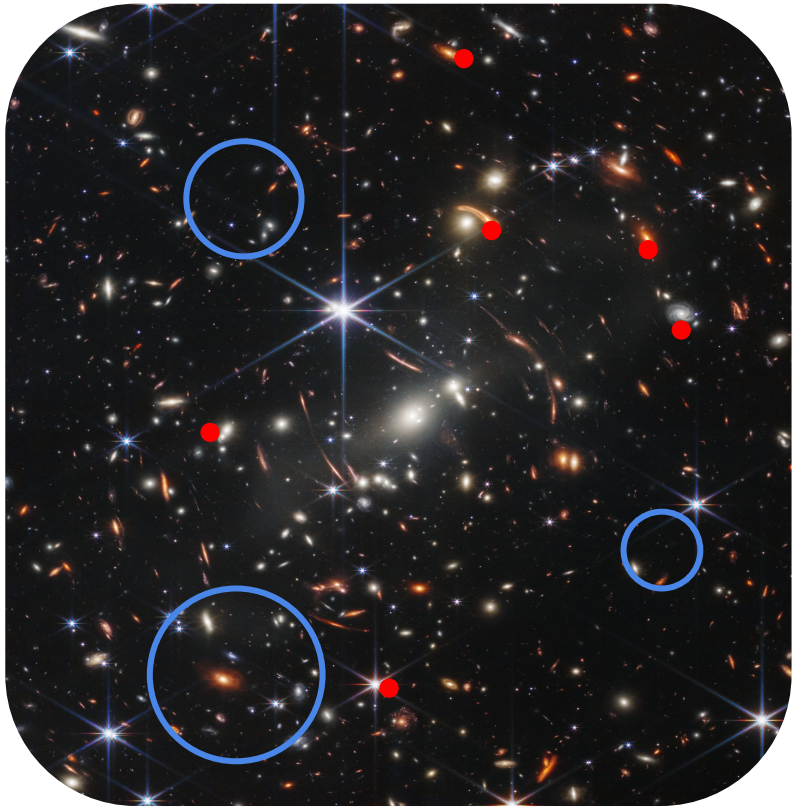
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2. Supernovae (SNe)



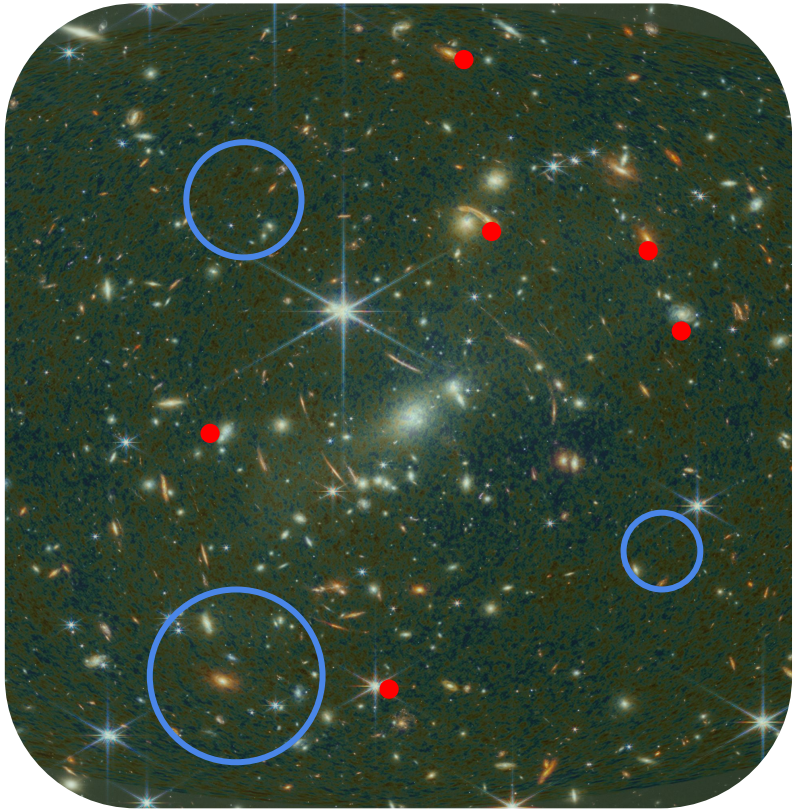
JWST Deep Field

The puzzling accelerated expansion of the Universe

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2. Supernovae (SNe)
3. Cosmic Microwave Background (CMB)



JWST Deep Field

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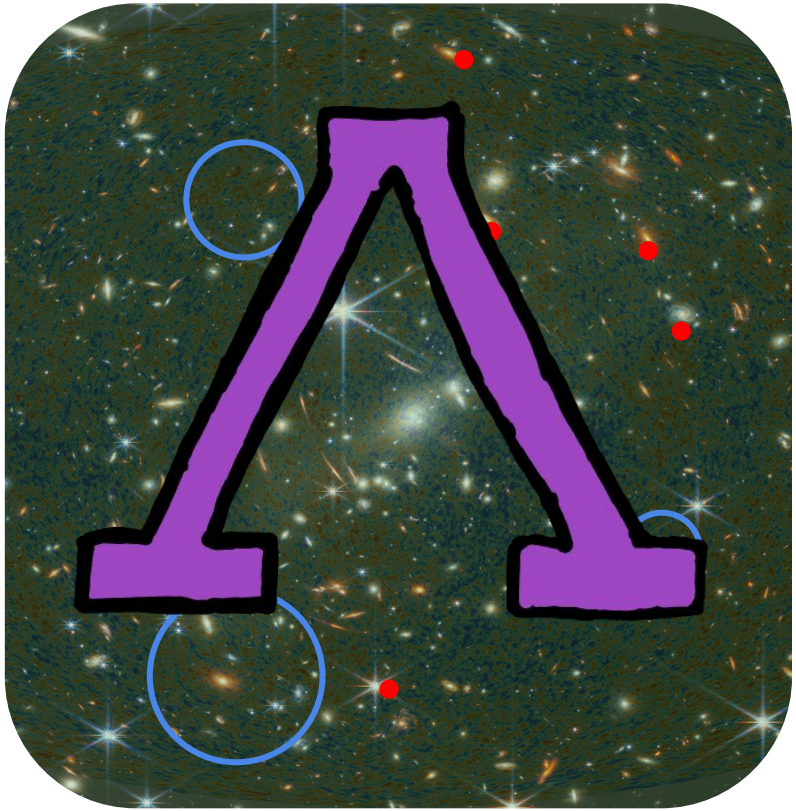
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1. Distribution of galaxies
2. Supernovae (SNe) → **1998**
3. Cosmic Microwave Background (CMB)

High-z Supernova Search Team (1998); Supernova Cosmology Project (1999)

~70% of the universe energy budget is in form of a vacuum-like component named **dark energy**

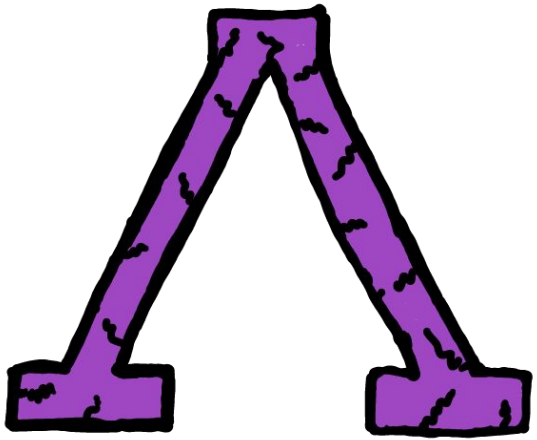


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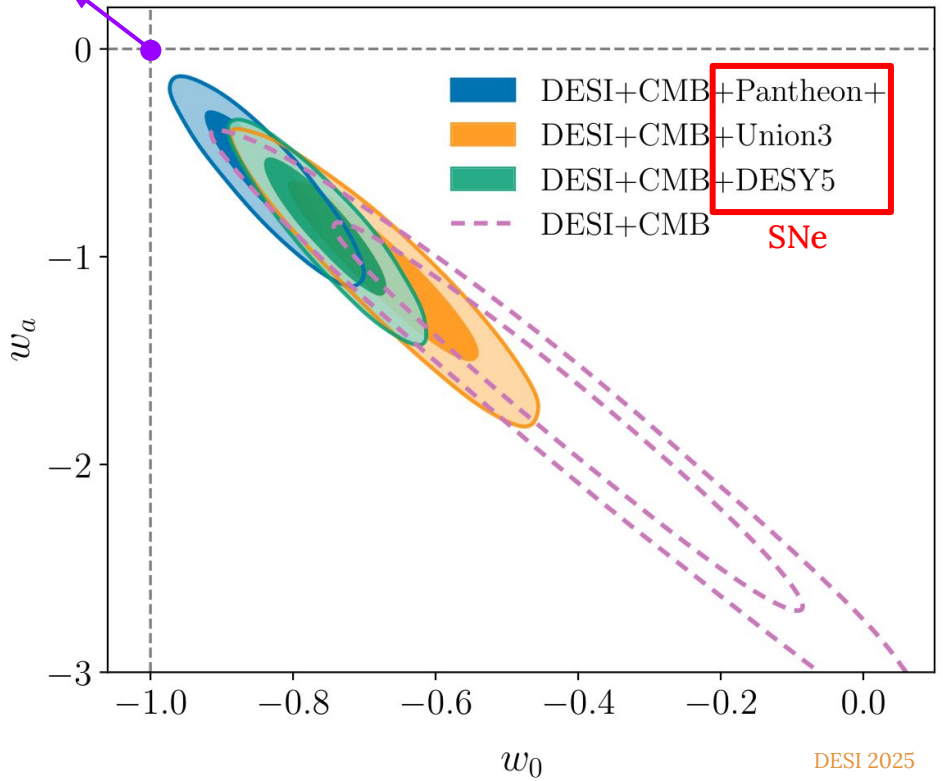
The puzzling accelerated expansion of the Universe

Recent data from the DESI collaboration indicate a **~3σ** preference for **evolving dark energy**

$$w_{\rm DE} \equiv p_{\rm DE}/\rho_{\rm DE}$$



$$w(a) = w_0 + (1-a)w_a$$

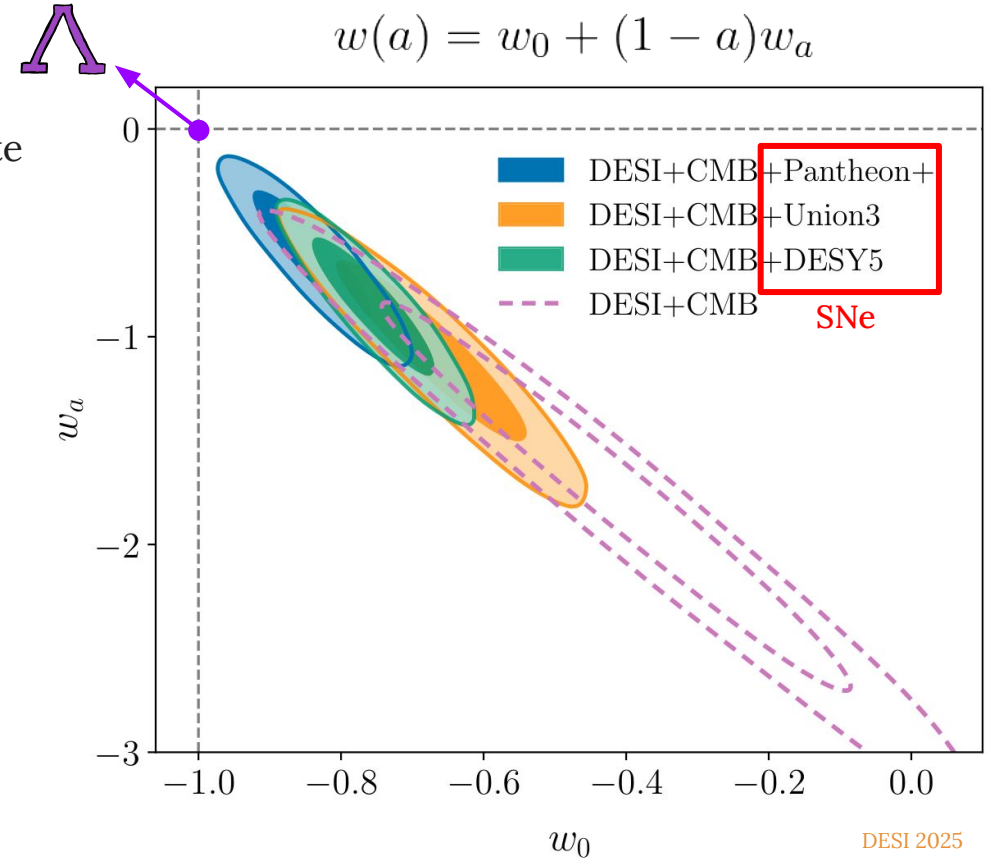


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DESI 2025

Outline

1. Geometric Probes of Dark Energy
2. Minimal Density Level Dark Energy Parametrization
3. Validation against CPL
4. Data Analysis
5. Summary & Future Directions

Geometric Probes of Dark Energy

constrain the **expansion history** of the universe through measurements of **distances at various redshifts**

$$\frac{H(z)}{H_0} = \left[\underbrace{\Omega_{bc}}_{\text{Baryons + DM}}(1+z)^3 + \underbrace{\Omega_\gamma}_{\text{Photons}}(1+z)^4 + \underbrace{\Omega_\nu}_{\text{Neutrinos}}\frac{\rho_\nu(z)}{\rho_{\nu,0}} + \underbrace{\Omega_{\rm DE}\, f_{\rm DE}(z)}_{\text{Dark Energy}}\right]^{1/2}$$

$$\Omega_{\rm DE} = 1 - \Omega_{bc} - \Omega_\gamma - \Omega_\nu$$

$$f_{\rm DE} \equiv \rho_{\rm DE}/\rho_{\rm DE,0}$$

Geometric Probes of Dark Energy

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CMB + BAO

Comoving Distance

$$D_M(z) = \int_0^z \frac{{\rm d}z'}{H(z')}$$

Hubble Distance

$$D_H(z) = 1/H(z)$$

Supernovae

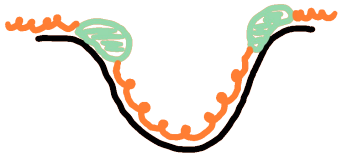
Luminosity Distance

$$d_L(z) = (1+z)\, D_M(z)$$

Geometric Probes of Dark Energy
Cosmic Sound Waves

- Gravity and pressure generated **sound waves** in the primordial plasma

$$\ddot{\delta}_\gamma - \underbrace{c_\gamma^2 \nabla^2 \delta_\gamma}_{\text{Photon Pressure}} = \underbrace{\nabla^2 \Psi_+}_{\text{Gravity}}$$



- After photon-decoupling, baryons no longer feel any pressure, and their **sound wave stalls**.



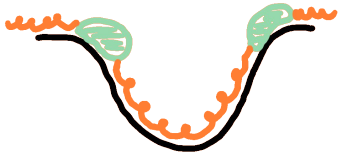
The **BAO** scale

Credit: Murali Saravanan

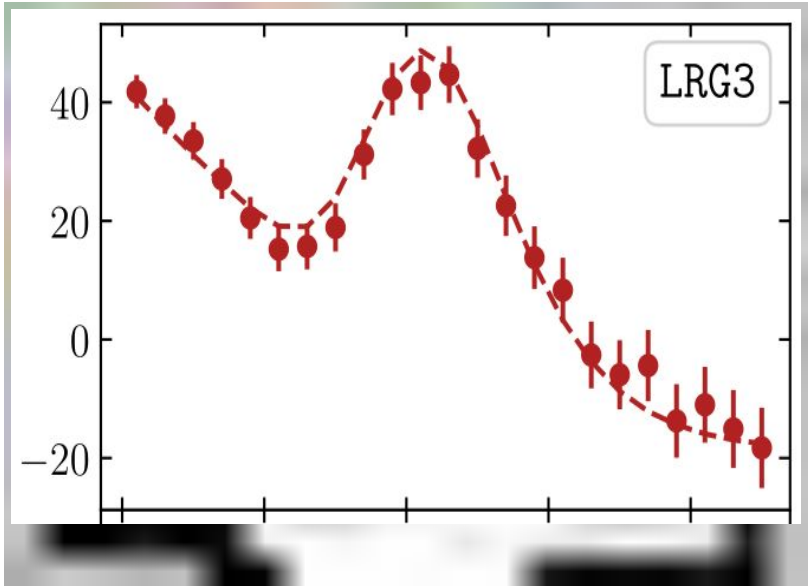
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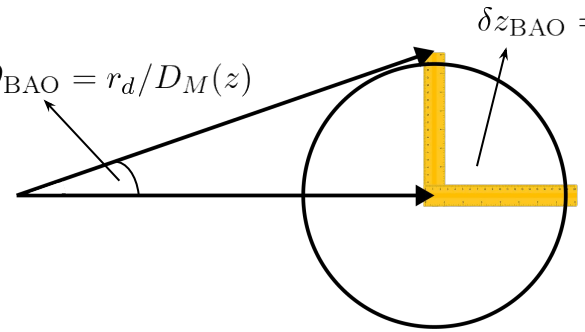
- After photon-decoupling, baryons no longer feel any pressure, and their **sound wave stalls**.
    - It imprints characteristic scale in the galaxy correlation function at **r_d~150 Mpc**



The **BAO** scale

Geometric Probes of Dark Energy
Cosmic Sound Waves





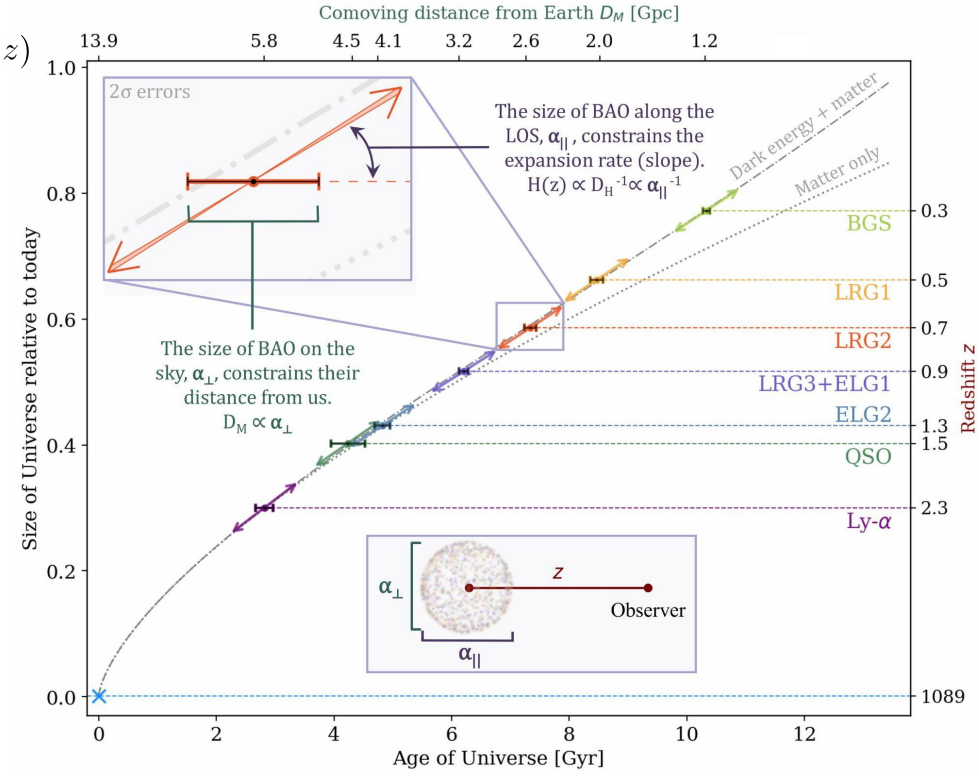
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- BAO surveys measure distances relative to the sound horizon



DESI DR2

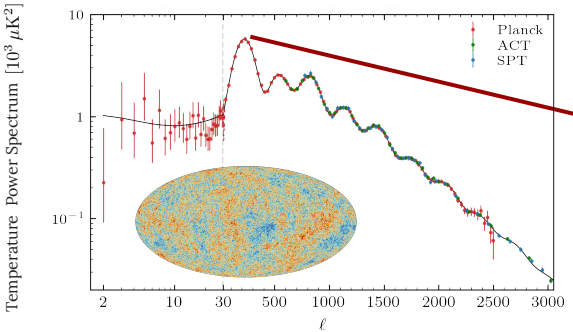
Geometric Probes of Dark Energy
Cosmic Sound Waves

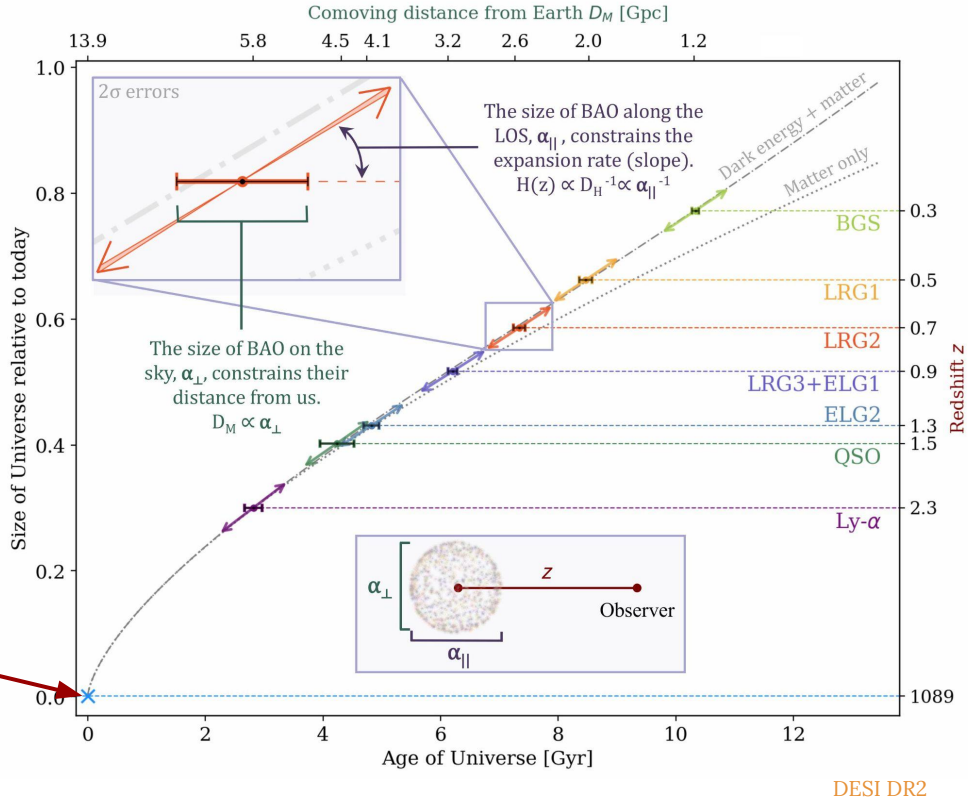
- BAO surveys measure distances relative to the sound horizon

$$\theta_{\rm BAO} = r_d/D_M(z) \qquad \delta z_{\rm BAO} = r_d/D_H(z)$$

- CMB provides the calibration
  - $\mathbf{r_d}$ from shape of CMB spectra (ω_{bc},ω_b)
  - $\boldsymbol{\theta_s}$ from first peak location

$$\theta_s = r_s(z_\star)/D_M(z_\star)$$





DESI DR2

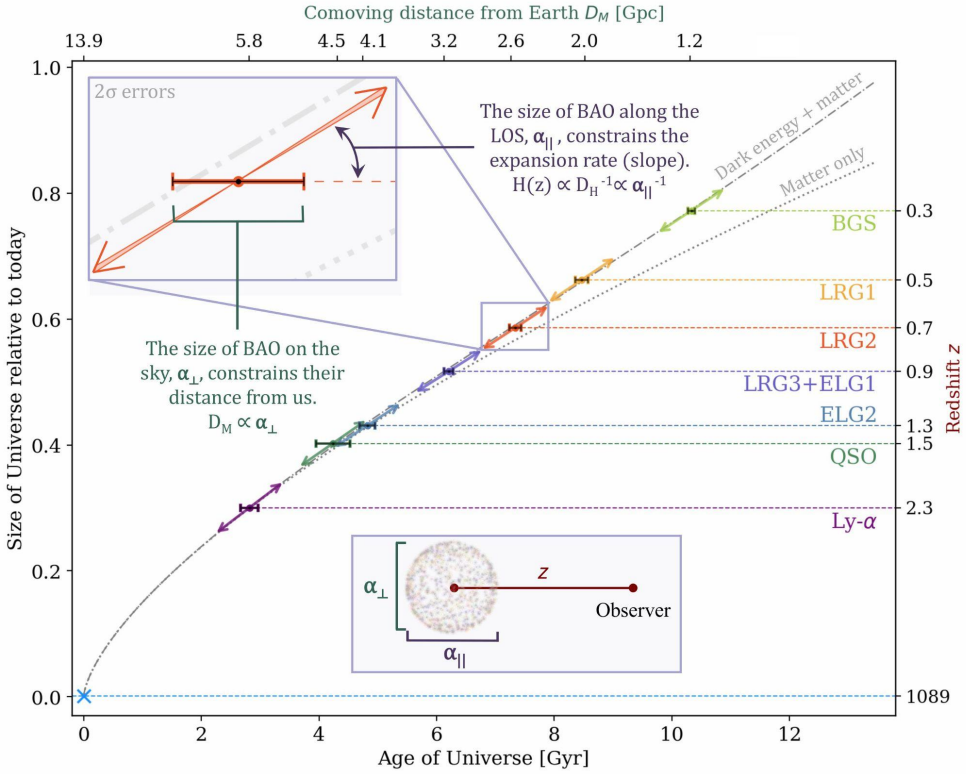
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Precise measurement of the variation of distances across the BAO redshift bins



DESI DR2

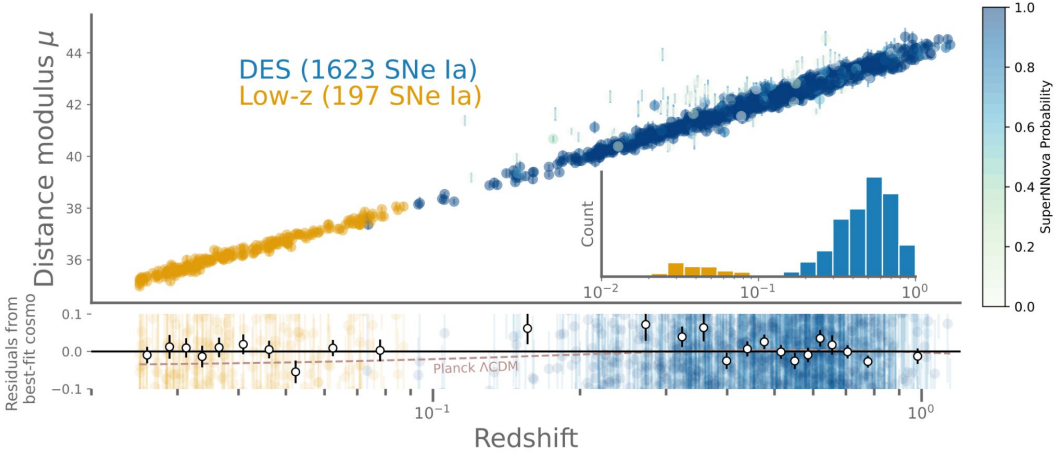
Geometric Probes of Dark Energy
Type IA Supernovae

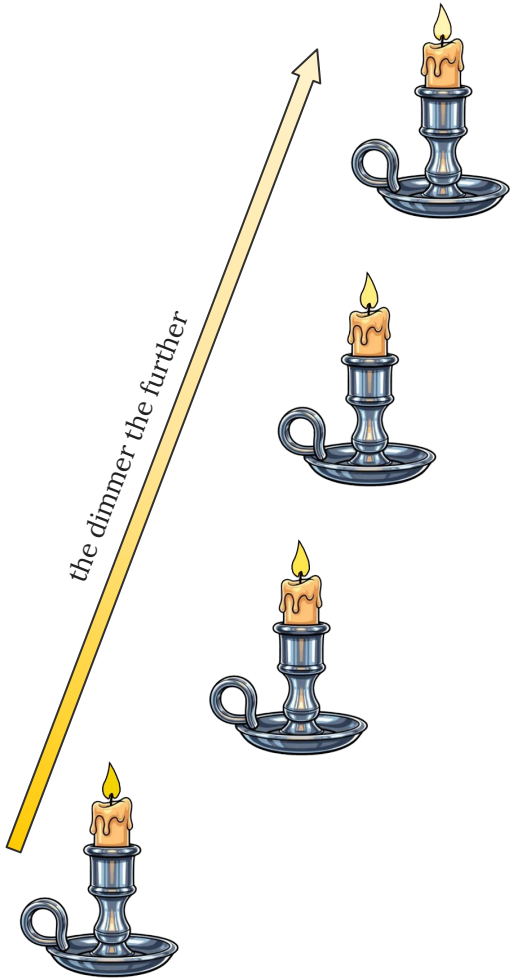
- **Standard candles:** uniform intrinsic luminosity (**M**) after light-curve calibration
    - Distance by comparing to the observed brightness (**m**)

$$m = M + 5\log_{10}\left[\frac{d_L(z)}{10\,\mathrm{pc}}\right]$$

Luminosity Distance

$$d_L(z) = (1+z)\,D_M(z)$$





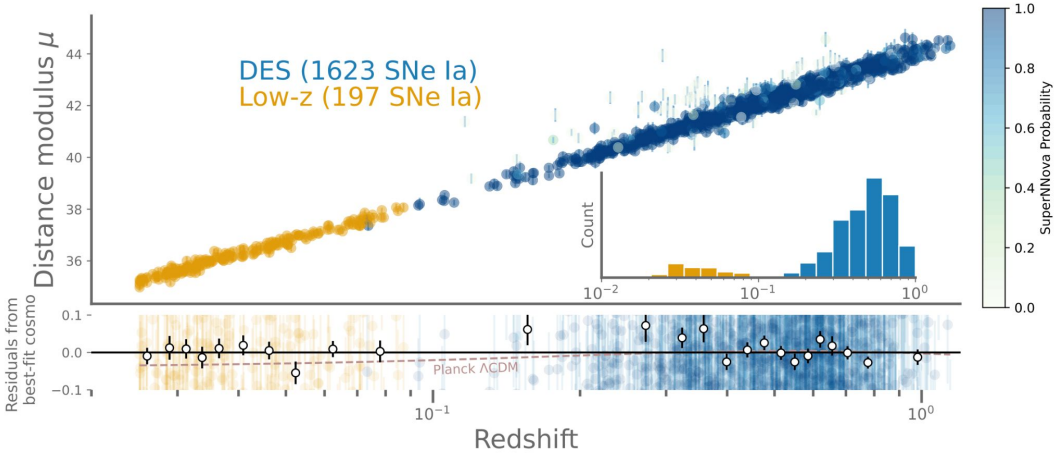
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Precise low-redshift measurement of Ω_m independent of acoustic physics

Geometric Probes of Dark Energy
Towards a Minimal Density Level Parametrization of DE

$$\frac{H(z)}{H_0} = \left[\Omega_{bc}\,(1+z)^3 + \Omega_\gamma\,(1+z)^4 + \Omega_\nu \frac{\rho_\nu(z)}{\rho_{\nu,0}} + \Omega_{\rm DE}\,\underbrace{f_{\rm DE}(z)}_{\text{Dark Energy}}\right]^{1/2}$$

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$$w_{\rm DE} \equiv p_{\rm DE}/\rho_{\rm DE}$$

- $\mathbf{f_{DE}}$ is the quantity that directly contributes to **H(z)**
- **w(z)** enters only through a nonlinear integral mapping

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- $\mathbf{f_{DE}}$ is the quantity that directly contributes to **H(z)**
- **w(z)** enters only through a nonlinear integral mapping
  - can amplify parameter uncertainties Maor, Brustein, Steinhardt 2000; Barger, Marfatia 2001; Wang, Freese 2004
  - introduce degeneracies and substantial prior dependence

Shlivko, Steinhardt 2024; Toomey, **Montefalcone,** McDonough, Freese 2025

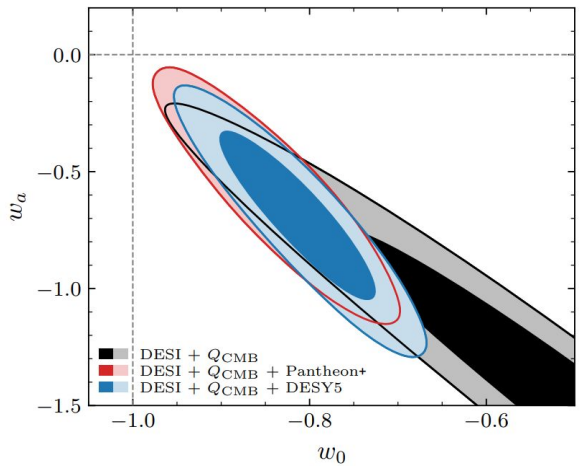
Geometric Probes of Dark Energy
Towards a Minimal Density Level Parametrization of DE

The choice of parametrization can impact the inferred evidence for evolving DE

DESI BAO + CMB

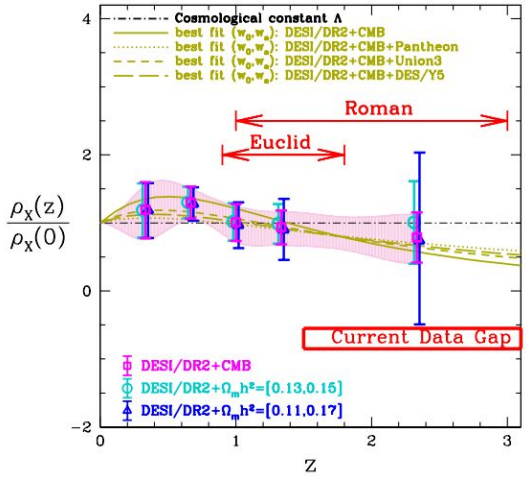
$$f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a\, z/(1+z)]$$

(w_0 ,w_a) parametrization **~2.5σ** from **Λ**



Montefalcone, Stiskalek 2026

Binned **f_{DE}** reconstruction **~1σ** from **Λ**



Wang, Freese 2025

A Minimal Density-Level Parametrization of DE

A two-parameter polynomial alternative to CPL

$$f_{\rm DE}(a) = 1 + f_a(1-a) + f_b(1-a)^2$$

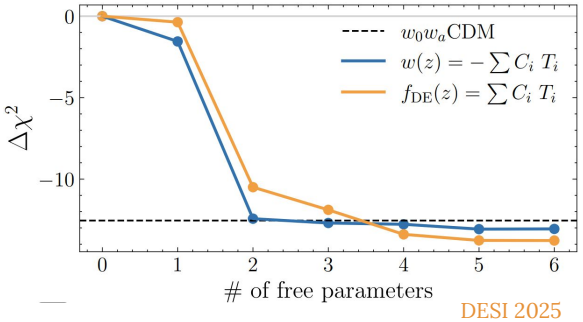
$f_a \equiv \mathrm{d}f_{\rm DE}/\mathrm{d}a\big|_{a=1}$ (linear rate of change of DE density)

$f_b \equiv \mathrm{d}^2 f_{\rm DE}/\mathrm{d}a^2\big|_{a=1}$ (curvature of DE evolution)

- Two DE degrees of freedom is all you need → avoids additional free-parameters from binned approaches

  Linder, Huterer 2005; DESI 2025

- Smooth analytical form allows natural inclusion of DE perturbations via **w(a)**



DESI 2025

$$w(a) = -1 + \frac{a\left[f_a + 2f_b(1-a)\right]}{3\left[1 + f_a(1-a) + f_b(1-a)^2\right]}\,.$$

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- Asymptotes to **Λ** for **a→0**
  - Consistent with many physically-motivated scenarios such as thawing quintessence
  - May limit applicability to scaling or freezing models
  - Avoids possible unphysical phantom behaviour of CPL $w_{\rm CPL}(a \to 0) = w_0 + w_a$

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- Asymptotes to **Λ** for **a→0**
  - Consistent with many physically-motivated scenarios such as thawing quintessence
  - May limit applicability to scaling or freezing models
  - Avoids possible unphysical phantom behaviour of CPL
- Allows $\mathbf{f_{DE}}$ to be negative
  - Encompasses a broader class of models than CPL, such as modified gravity or interacting dark-sector scenarios
  - Results in a singularity in **w(a)** to deal with

A Minimal Density-Level Parametrization of DE

It's all about the Pivot

$$f_{\rm DE}(a) = 1 + f_a(1-a) + f_b(1-a)^2$$

$$f_p \equiv f_{\rm DE}(a_p) \qquad w_p \equiv w(a_p)$$

$$f_{\rm DE}(z) = f_p - 3\,(1+w_p)\,f_p\left(\frac{z_p - z}{1+z}\right) + \frac{1 - f_p + 3\,(1+w_p)\,f_p\,z_p}{z_p^2}\left(\frac{z_p - z}{1+z}\right)^2$$

- Transparent characterization of DE evolution at the epoch ($\mathbf{z_p}$) where data are most sensitive
- **($\mathbf{w_p}$, $\mathbf{f_p}$)** are decorrelated and constrained by a distinct subset of the data

A Minimal Density-Level Parametrization of DE

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CMB + BAO: Precise measurement of the variation of distances across the BAO redshift bins

SNe: Precise low-redshift measurement of $\mathbf{\Omega_m}$ independent of acoustic physics

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CMB + BAO: Precise measurement of the variation of distances across the BAO redshift bins → DE rate of change $\left.\frac{d\ln f_{\rm DE}}{d\ln(1+z)}\right|_{z_p} = 3(1+w_p)$

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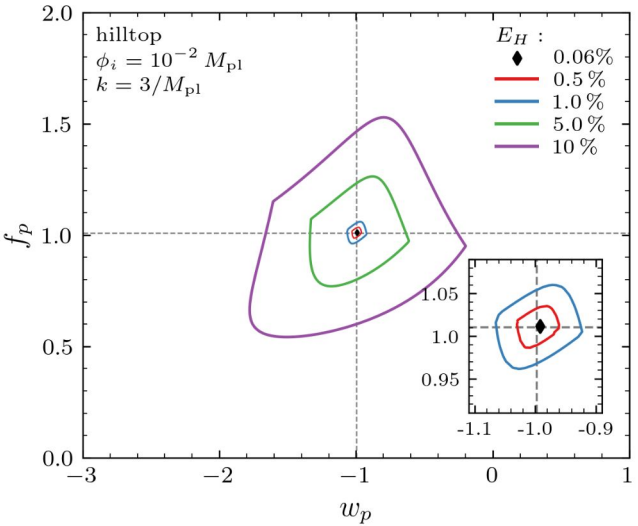
SNe: Precise low-redshift measurement of $\mathbf{\Omega_m}$ independent of acoustic physics → DE normalization $f_{\rm DE}(z_p) = f_p$

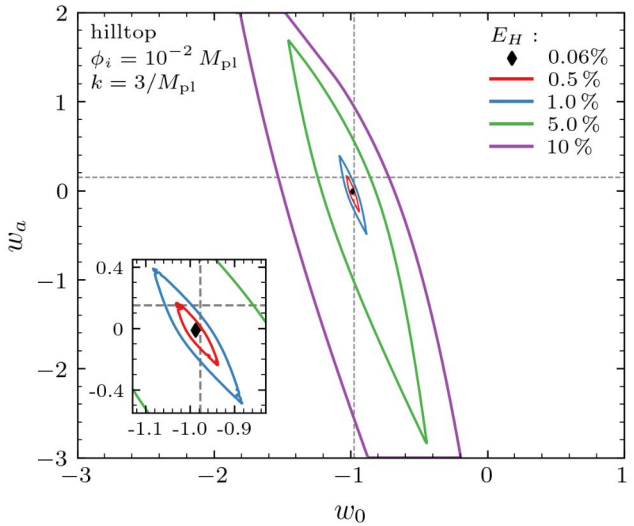
Validation against CPL

Expansion Rate of Benchmark Quintessence

$$V_{\rm hill}(\phi) = V_0\left(1 - \frac{k^2\phi^2}{2}\right)$$

$$E_H = \max_{z<4}\left|\frac{H_{\rm fit}(z) - H_\phi(z)}{H_\phi(z)}\right|$$





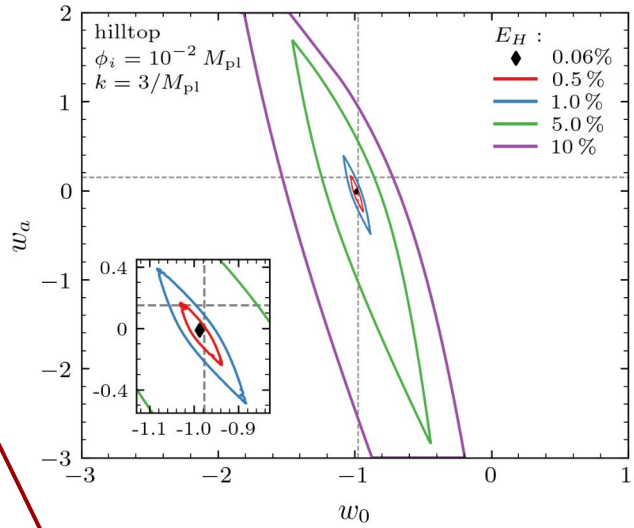
Contours	compact and orthogonal	elongated along $\Delta w_a/\Delta w_0 \cong -5$, weak sensitivity to w_a
Best-fits	recovers true values	true values outside 1% error contour

Validation against CPL

Expansion Rate of Benchmark Quintessence

This has implications for prior dependence:

- many combinations of **(w_0, w_a)** produce a similar expansion history



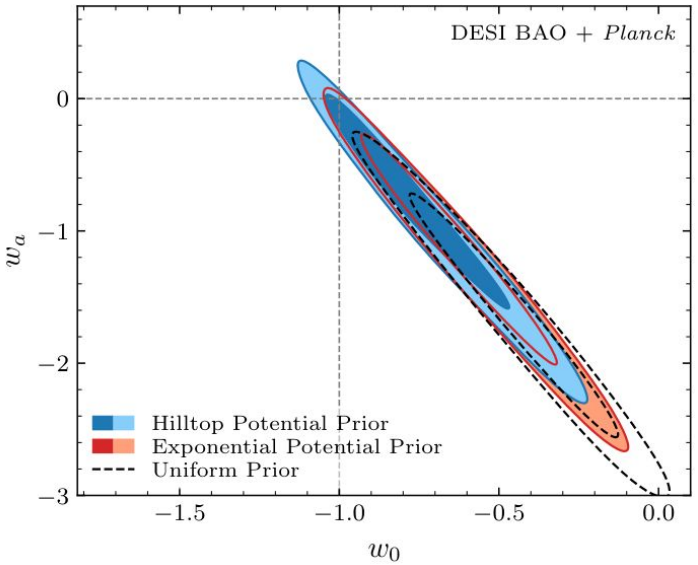
Contours	compact and orthogonal	elongated along $\Delta w_a/\Delta w_0 \cong -5$, weak sensitivity to w_a
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Validation against CPL

Expansion Rate of Benchmark Quintessence

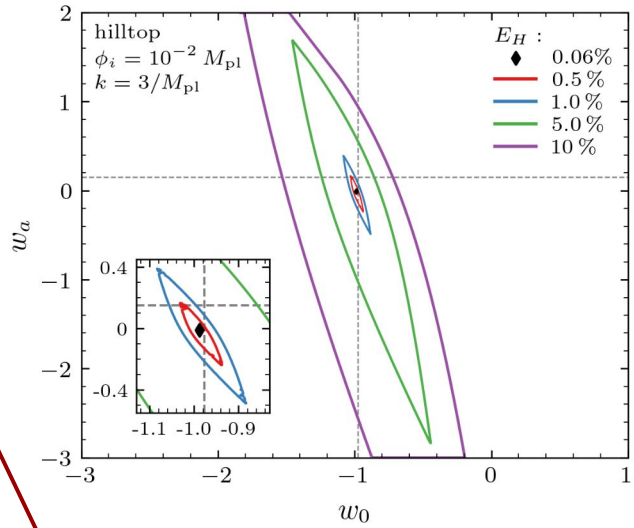
This has implications for prior dependence:

- many combinations of (w_0, w_a) produce a similar expansion history



~2σ shift from theory-informed prior

Toomey, **Montefalcone**, McDonough, Freese 2025



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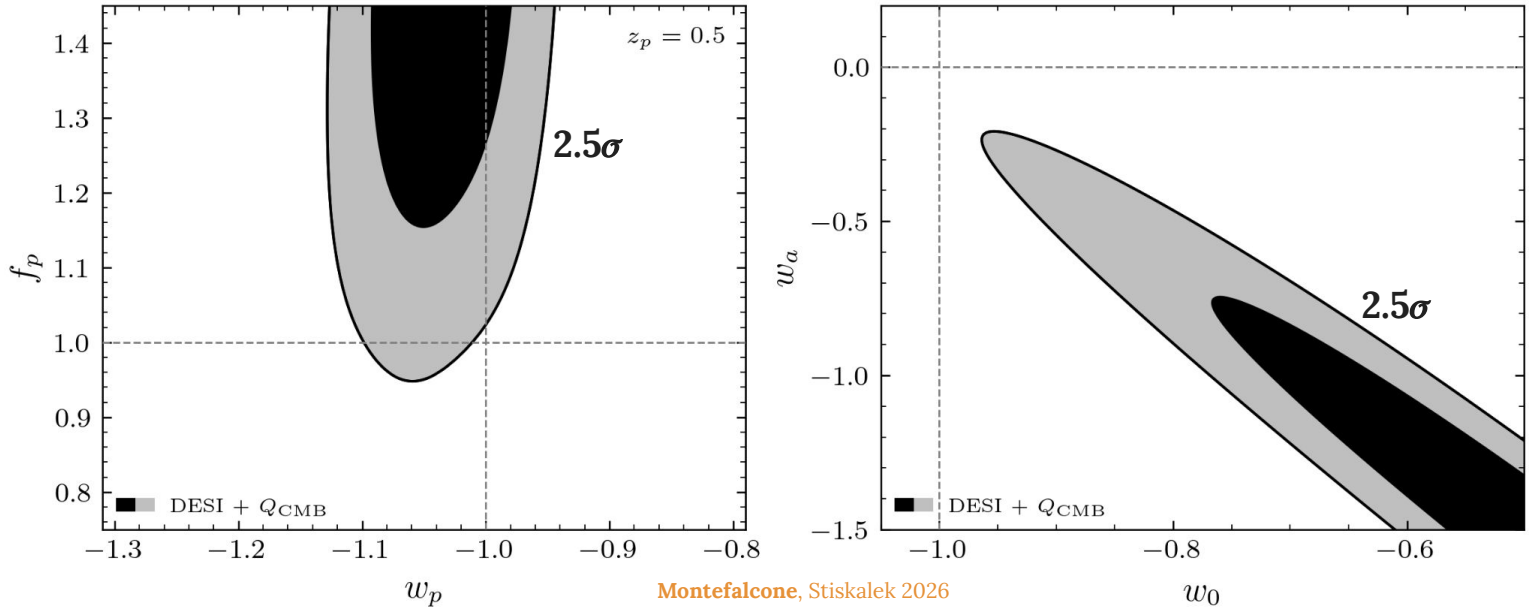
true values outside 1% error contour

Constraints on Evolving Dark Energy
Compressed CMB + DESI BAO

- **percent** level constraint on w_p
- f_p is poorly determined

$$w_p = -1.03 \pm 0.04$$
$$f_p = 1.66^{+0.41}_{-0.23}$$

Q_{CMB} : Correlated Gaussian prior on $(\omega_b, \omega_{bc}, \theta_s)$



Constraints on Evolving Dark Energy
Compressed CMB + DESI BAO + SNe

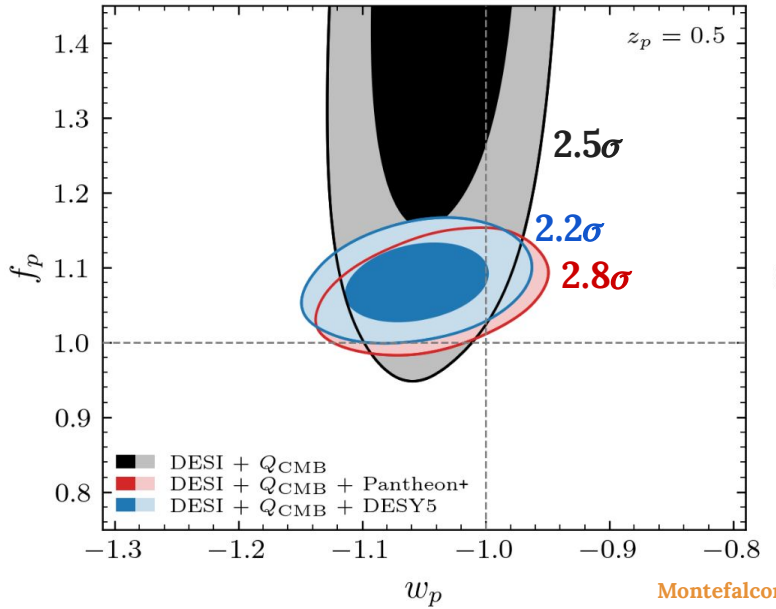
- **percent** level constraint on w_p
- f_p is pinned down by SNe data at the **percent** level

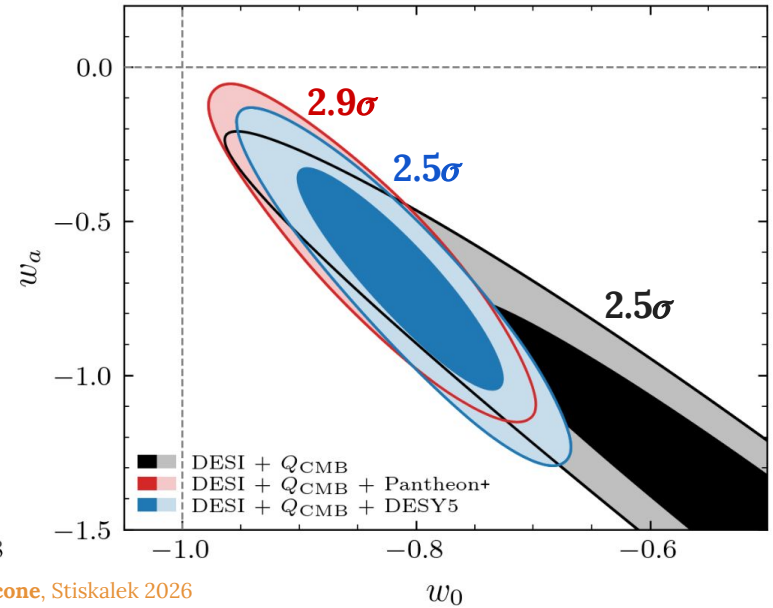
$$w_p = -1.03 \pm 0.04$$
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$$w_p = -1.04 \pm 0.04$$
$$f_p = 1.07 \pm 0.04$$

$$w_p = -1.06 \pm 0.04$$
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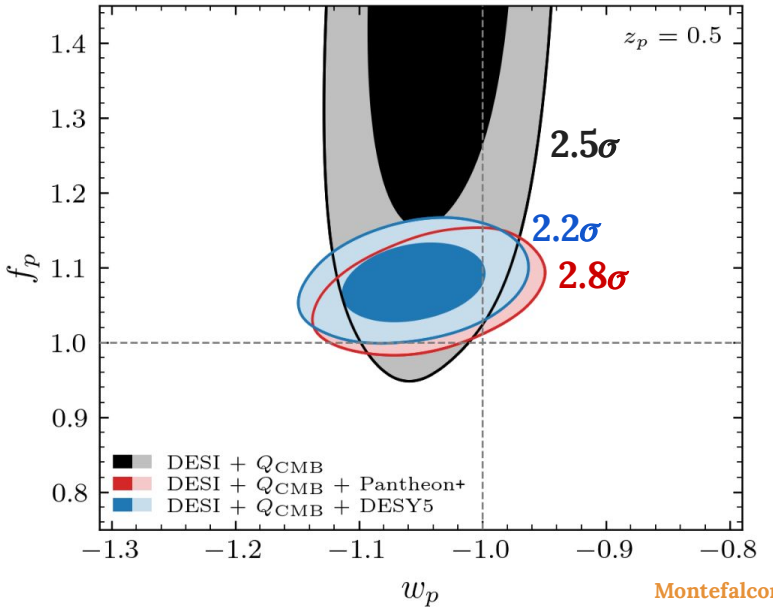
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Compressed CMB + DESI BAO + SNe

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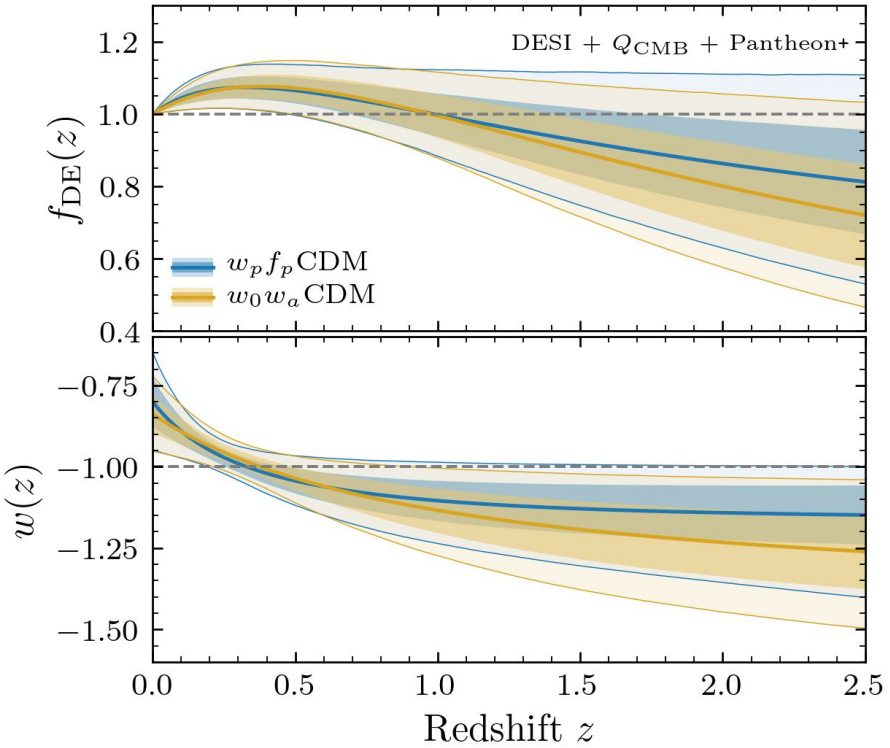
$w_p = -1.06 \pm 0.04$
$f_p = 1.08 \pm 0.04$



- consistent preference for evolving DE at the **~2σ** level
- parameters are decorrelated and aligned with distinct observables
- Coincidence: why are $\mathbf{f_p}$ and $\mathbf{w_p}$ close to their **Λ** limit?

Constraints on Evolving Dark Energy
Compressed CMB + DESI BAO + SNe

- Consistent reconstruction with CPL
- Marginal hint of phantom behaviour
  - mildly less pronounced than CPL



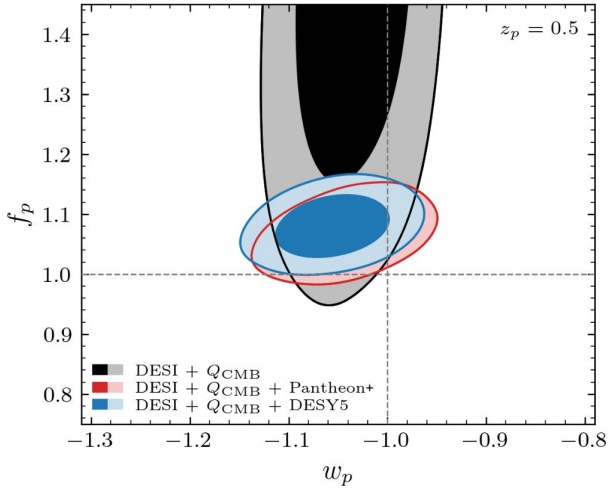
Summary

- A **two-parameter density-level** polynomial parametrization of DE as a **complementary** alternative to the CPL prescription
- An **immediate and transparent** characterization of the state of DE via **(w_p, f_p)** at the redshift where data are most constraining
- **Percent-level measurements** of both DE parameters individually circumventing the **intrinsic redundancies** of the **(w_0, w_a)** basis
  - CMB+BAO→ **w_p**; SNe→ **f_p**
  - **~2σ** preference for dynamical DE, consistent with CPL
  - Proximity of both **(w_p, f_p)** to their **Λ** limit reinforces the case for **caution** in interpreting the current evidence for dynamical DE

Future Directions

- Incorporate cosmological **perturbations** in the analysis and **forecast** the reach of next-generation BAO, SNe, and CMB surveys
- A dedicated **prior-sensitivity** study
- Validate the **(w_p, f_p)** formulation against a broader spectrum of evolving DE scenarios

Grazie per l'attenzione!



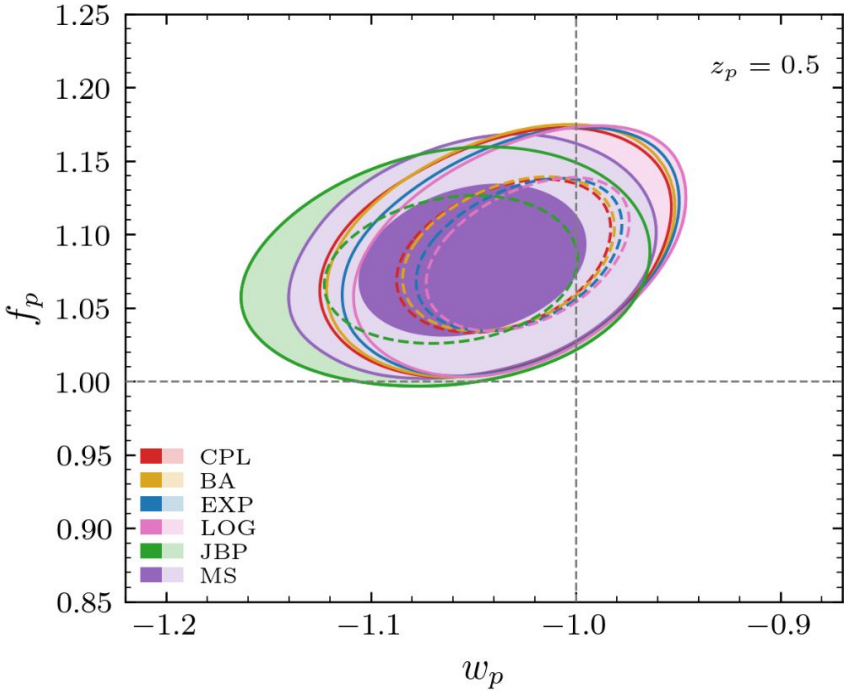
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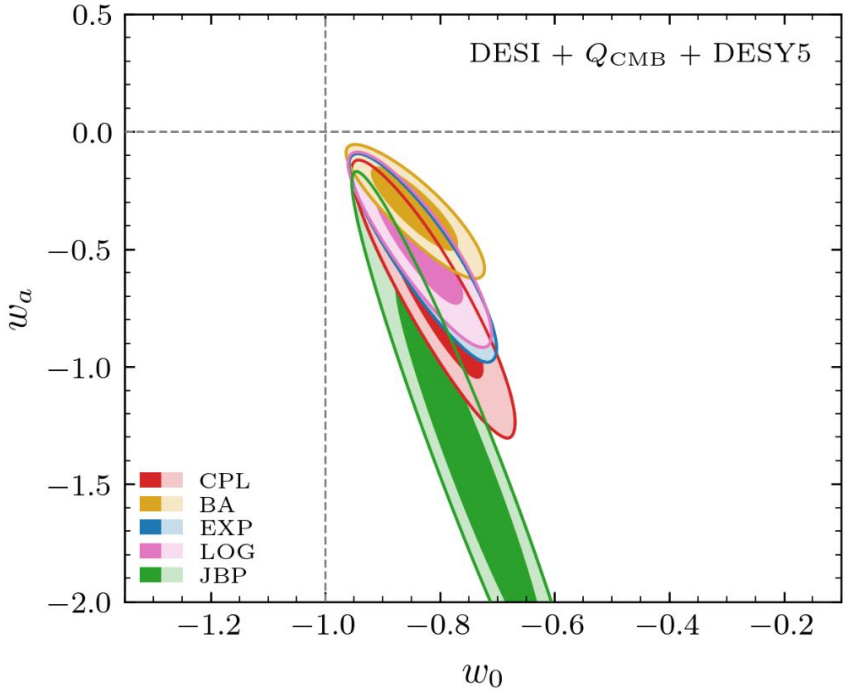
$w_p = -1.04 \pm 0.04$
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It is all about the pivot!

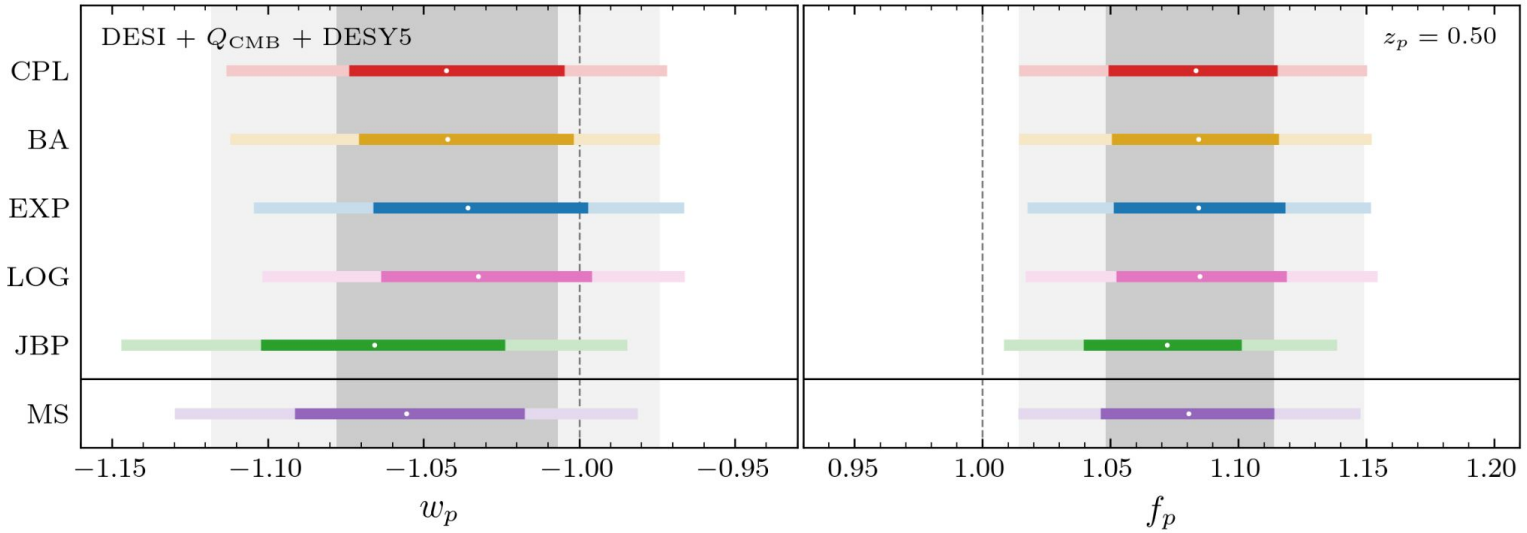
Param.	Functional Form
BA	$w_0 + w_a \frac{1-a}{a^2+(1-a)^2}$
EXP	$(w_0 - w_a) + w_a \exp(1-a)$
LOG	$w_0 - w_a \ln a$
JBP	$w_0 + w_a\, a(1-a)$
CPL	$w_0 + w_a(1-a)$





It is all about the pivot!

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Back-up Slides

Gabriele Montefalcone

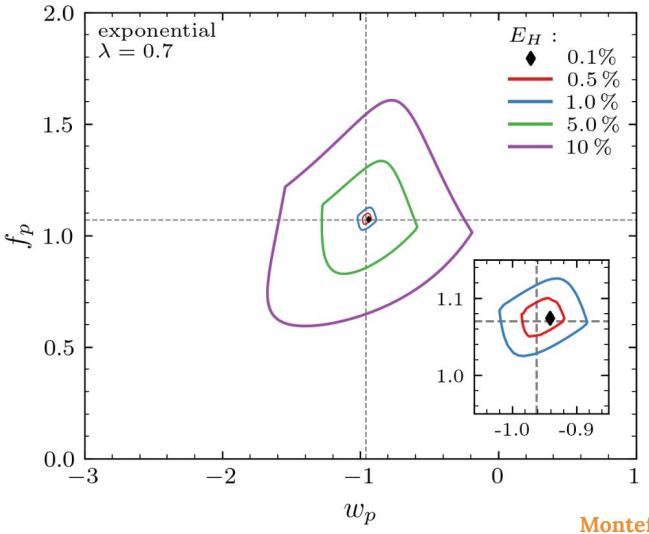
Weinberg Institute for Theoretical Physics, University of Texas at Austin

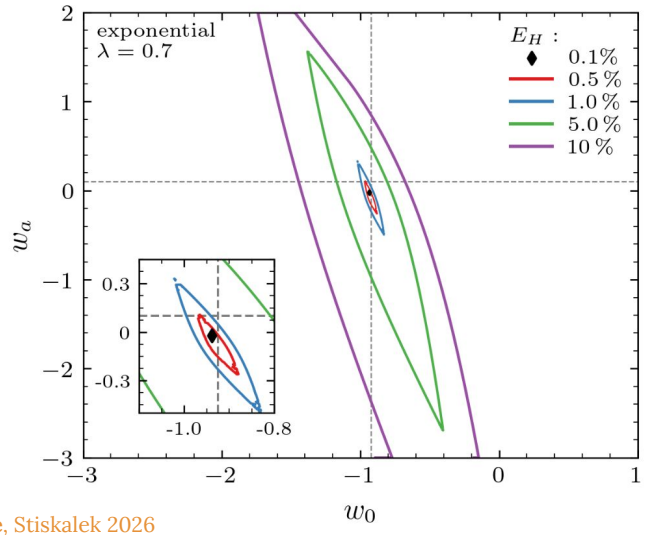
Validation against CPL

Expansion Rate of Benchmark Quintessence

$$V_{\rm exp}(\phi) = V_0 e^{\lambda\phi/M_{\rm pl}}$$

$$E_H = \max_{z<4} \left| \frac{H_{\rm fit}(z) - H_\phi(z)}{H_\phi(z)} \right|$$





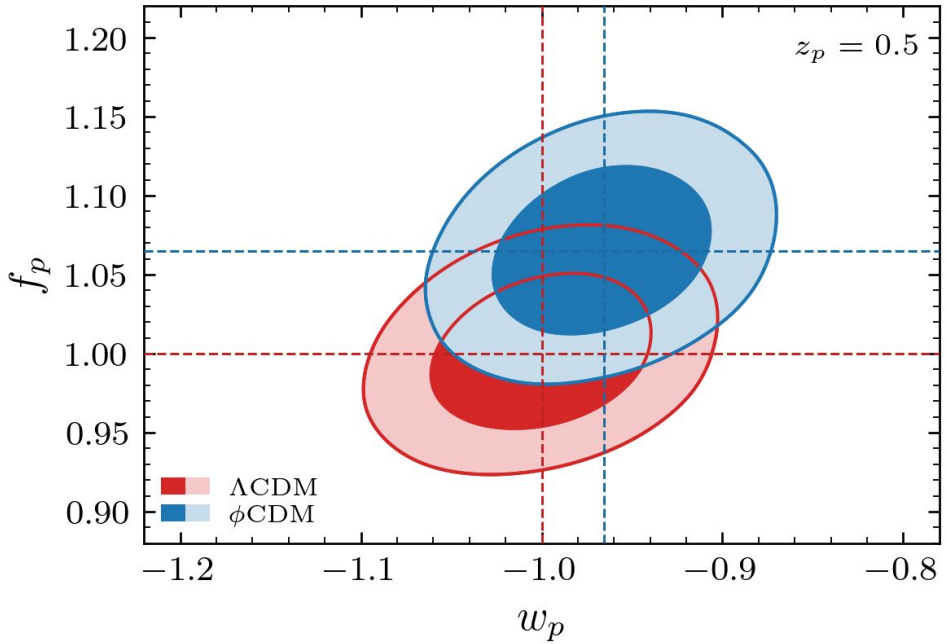
Montefalcone, Stiskalek 2026

Contours	compact and orthogonal	elongated along $\Delta w_a/\Delta w_0 \cong -5$, weak sensitivity to w_a
Best-fits	recovers true values	true values outside 1% error contour

Validation of Density-Level Parametrization of DE

Warm-up with Mock data

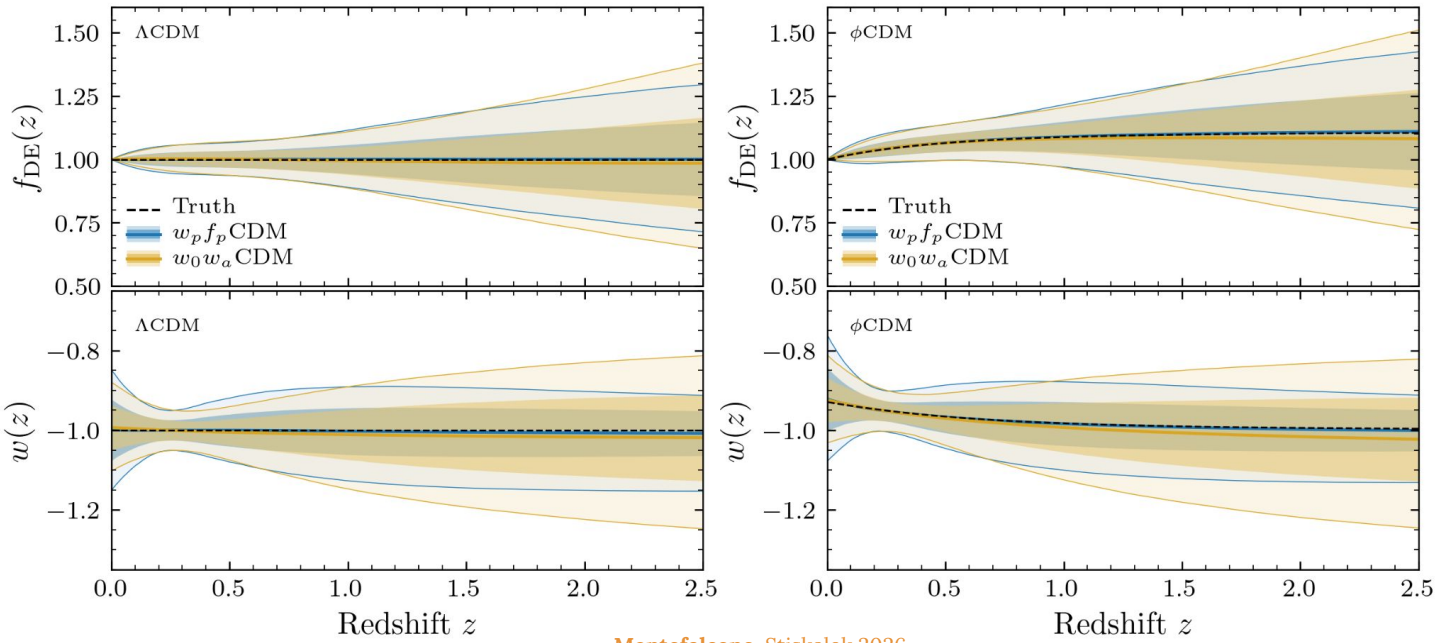
- Recovers underlying DE dynamic
- Both parameters are well-measured and decorrelated



Validation of Density-Level Parametrization of DE

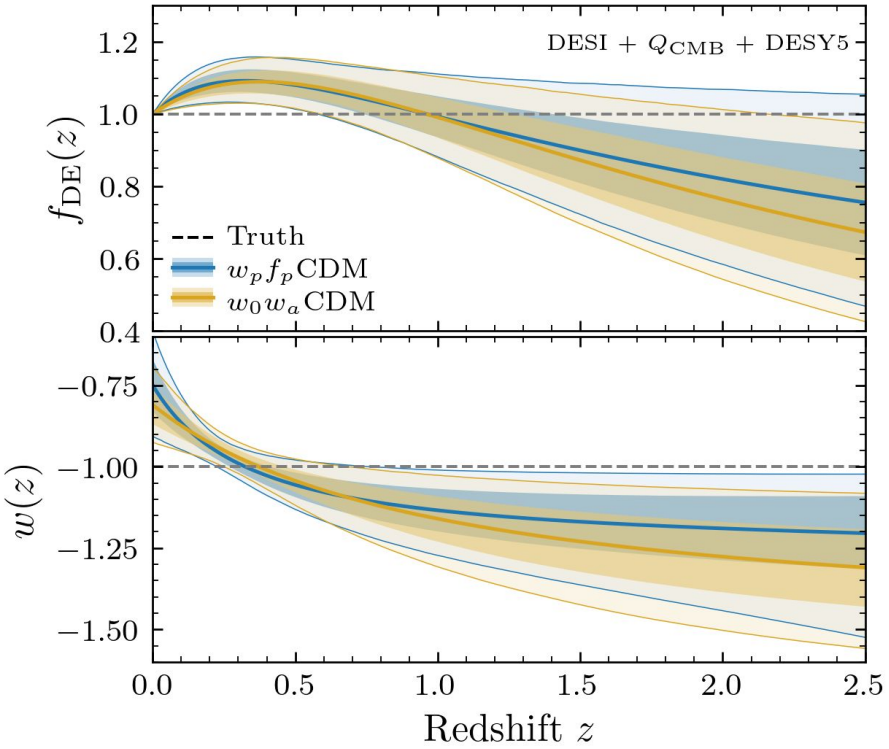
Warm-up with Mock data

CPL uncertainty bands become visibly broader at higher redshifts



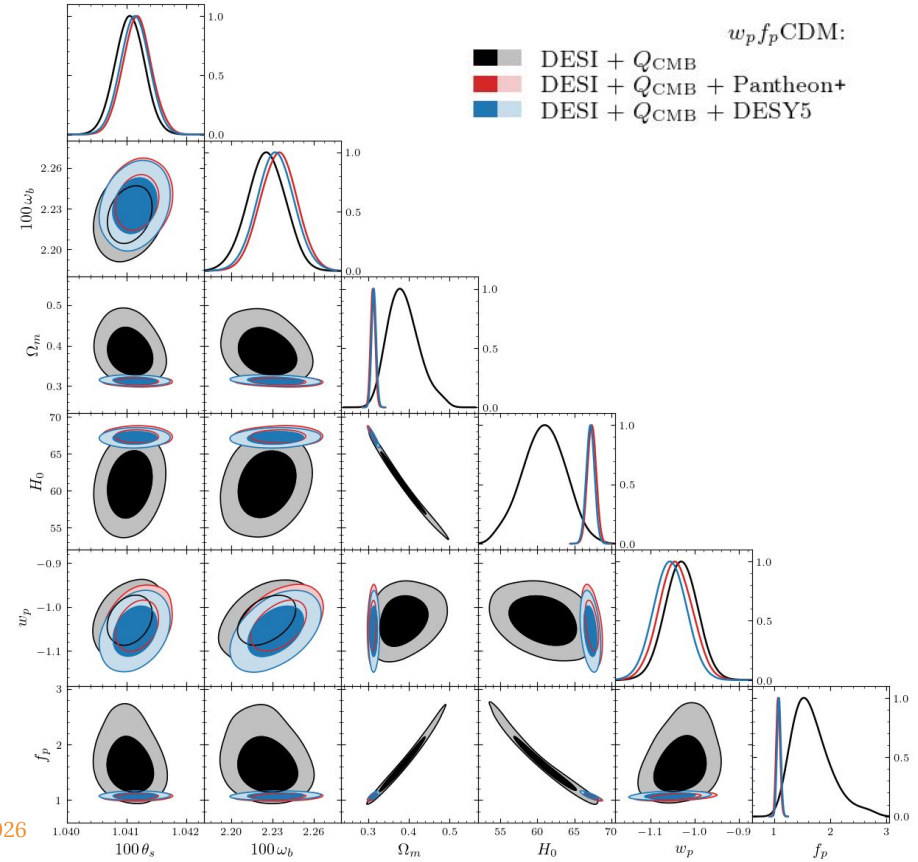
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Compressed CMB + DESI BAO + SNe

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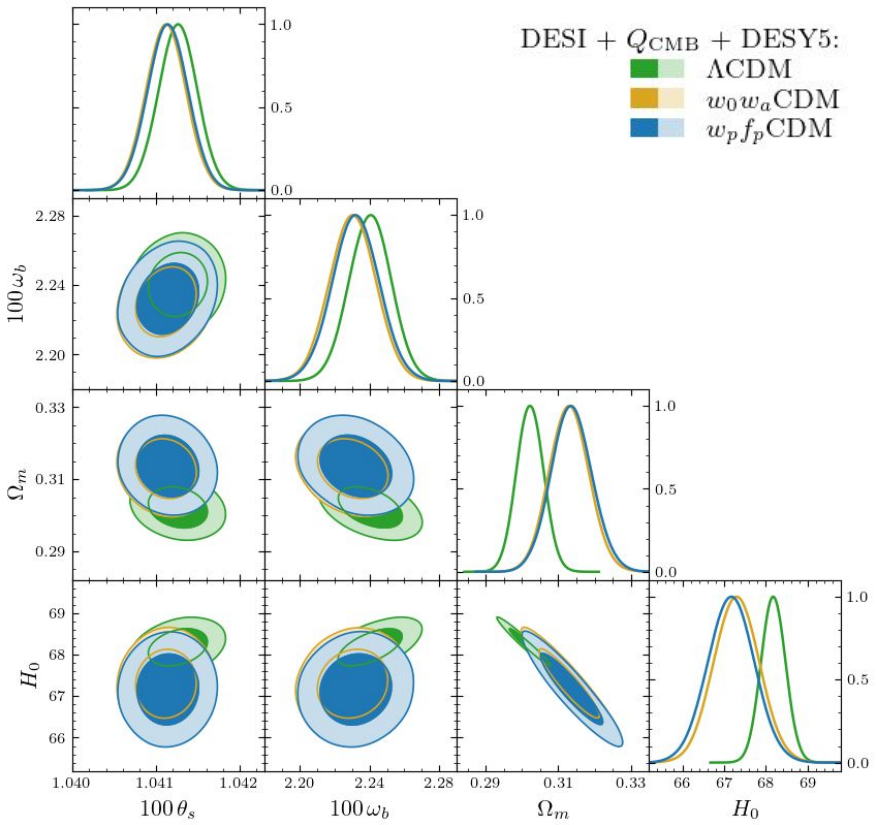
Constraints on Evolving Dark Energy
Additional Triangle Plots

- SNe break $(\mathbf{f_p}, \mathbf{H_0}, \boldsymbol{\Omega_m})$ degeneracy through $\boldsymbol{\Omega_m}$ measurement independent of r_d



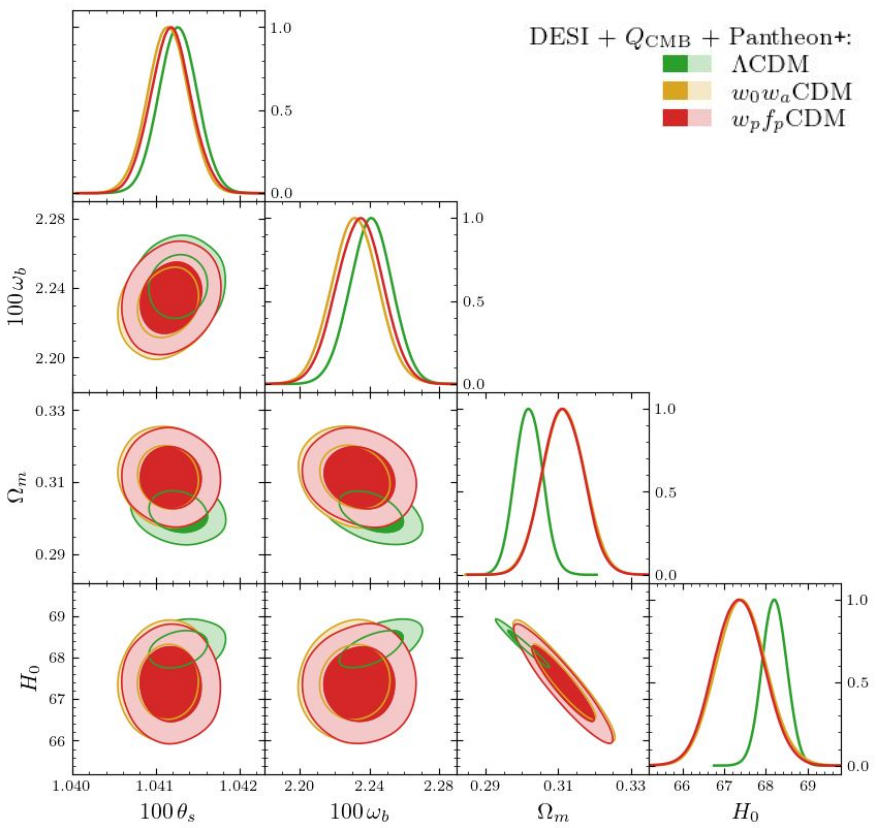
Constraints on Evolving Dark Energy

Additional Triangle Plots



Constraints on Evolving Dark Energy

Additional Triangle Plots



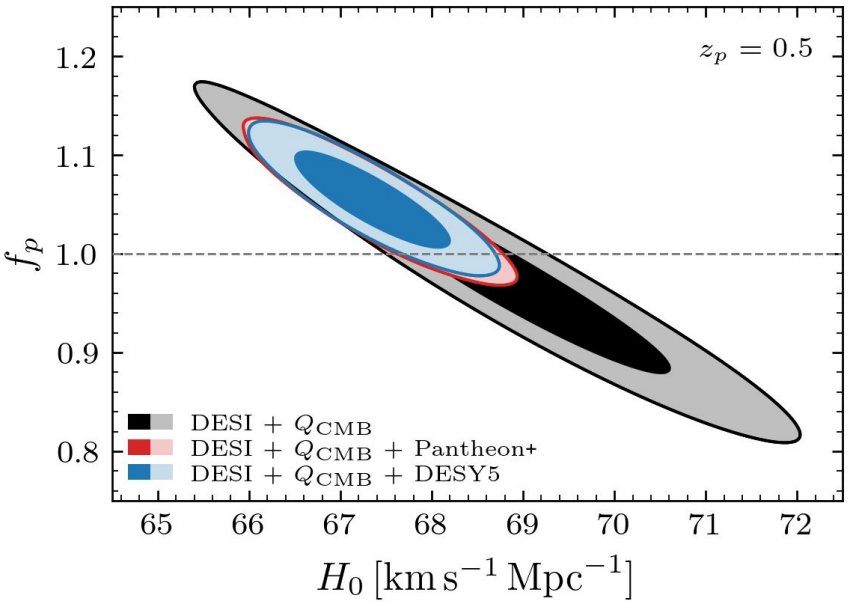
Constraints on Evolving Dark Energy

Single f_p Parametrization

$$f_{\rm DE}(z) = 1 + (f_p - 1)\,\frac{z\,(1+z_p)}{z_p\,(1+z)}$$

$$f_p = \begin{cases} 0.99^{+0.05}_{-0.08} & \text{DESI} + Q_{\rm CMB}\,, \\ 1.05 \pm 0.04 & \text{DESI} + Q_{\rm CMB} + \text{Pantheon+}\,, \\ 1.05 \pm 0.03 & \text{DESI} + Q_{\rm CMB} + \text{DESY5}\,. \end{cases}$$

- Consistent with **Λ** at the **1σ** level
- A single degree of freedom can dilute the preference for evolving DE



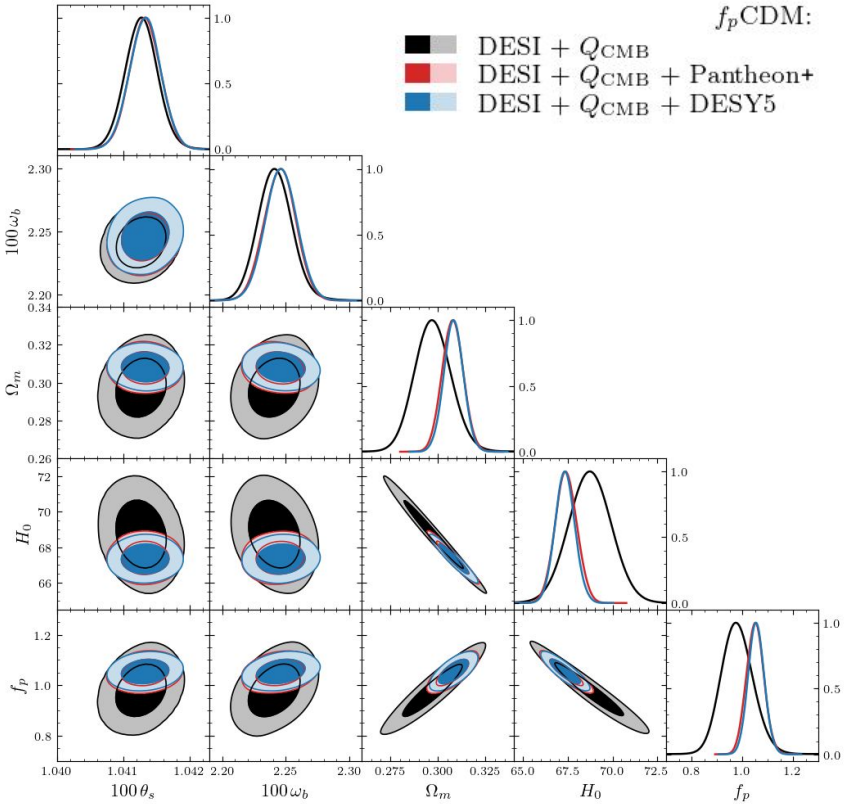
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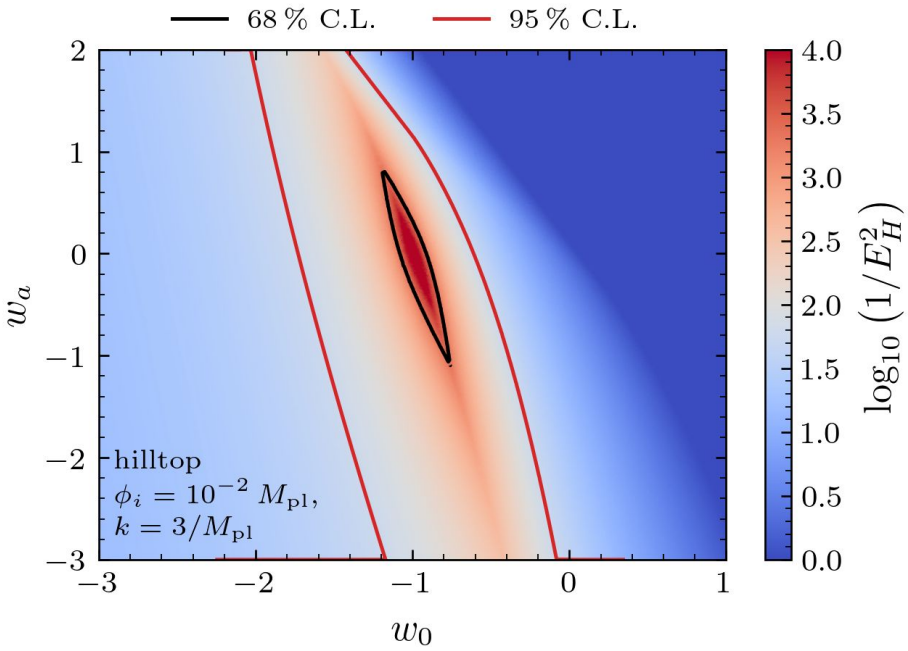
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Theory-Informed Priors
From Quintessence to CPL

$$V_{\rm hill}(\phi) = V_0\left(1 - \frac{k^2\phi^2}{2}\right)$$

$$E_H = \max_{z<4}\left|\frac{H_{\rm fit}(z) - H_\phi(z)}{H_\phi(z)}\right|$$

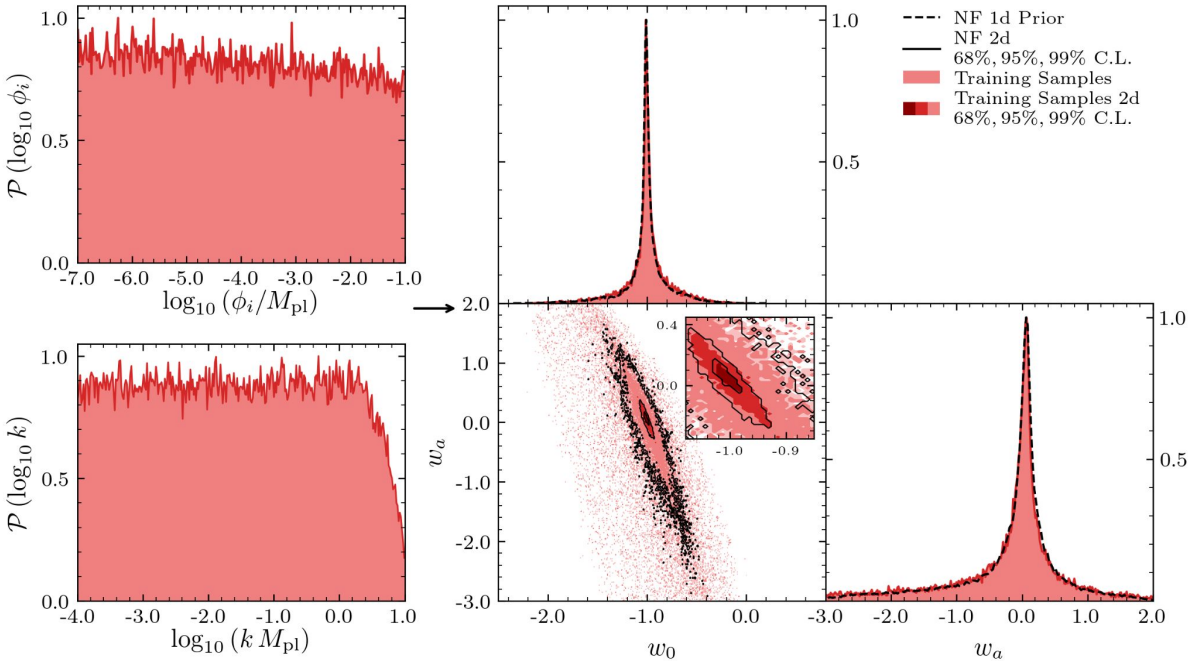


Toomey, **Montefalcone**, McDonough, Freese 2025

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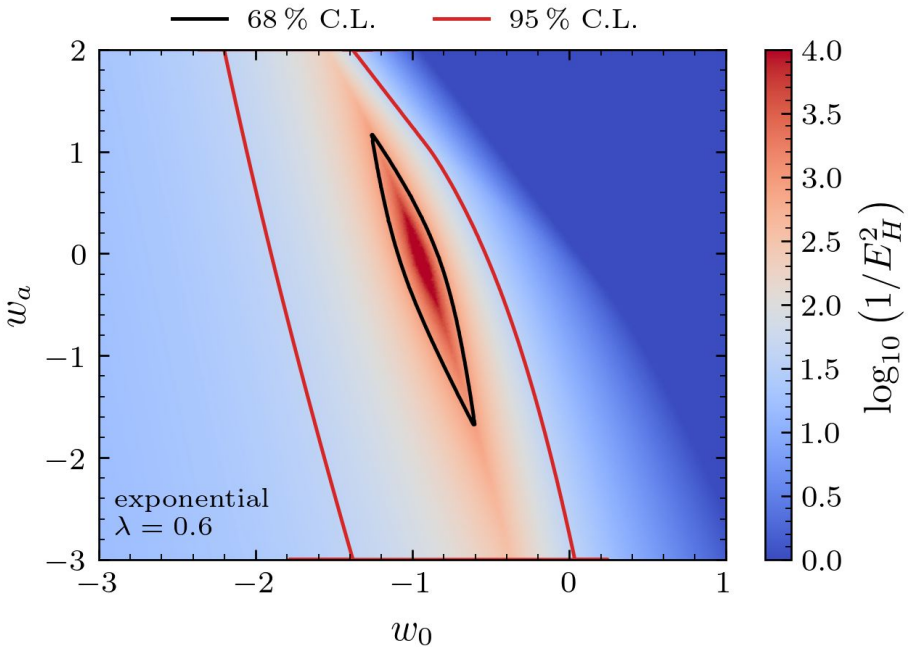


Toomey, **Montefalcone**, McDonough, Freese 2025

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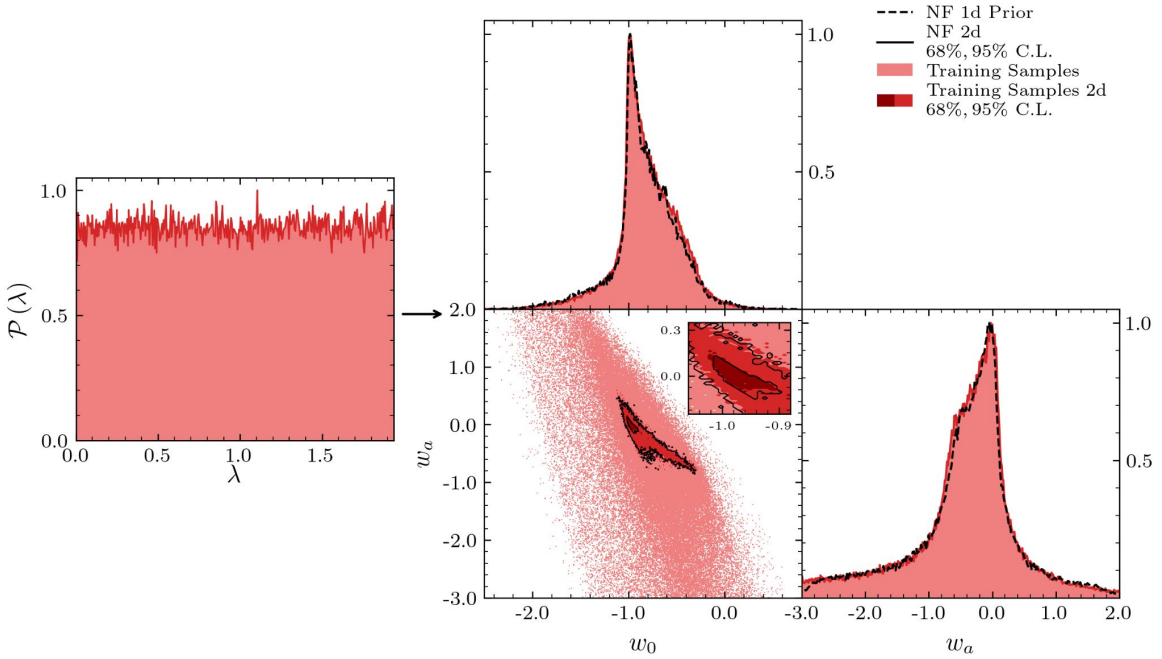


Toomey, **Montefalcone,** McDonough, Freese 2025

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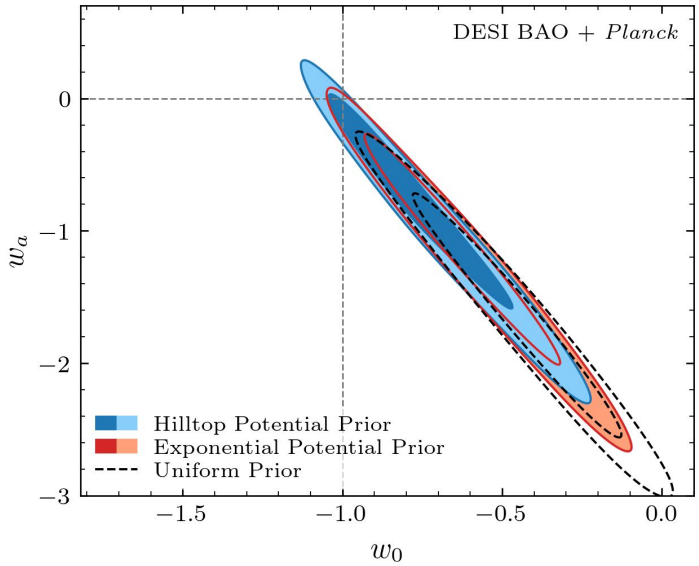
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Toomey, **Montefalcone**, McDonough, Freese 2025

Theory-Informed Priors
Impact on the Preference for Evolving DE

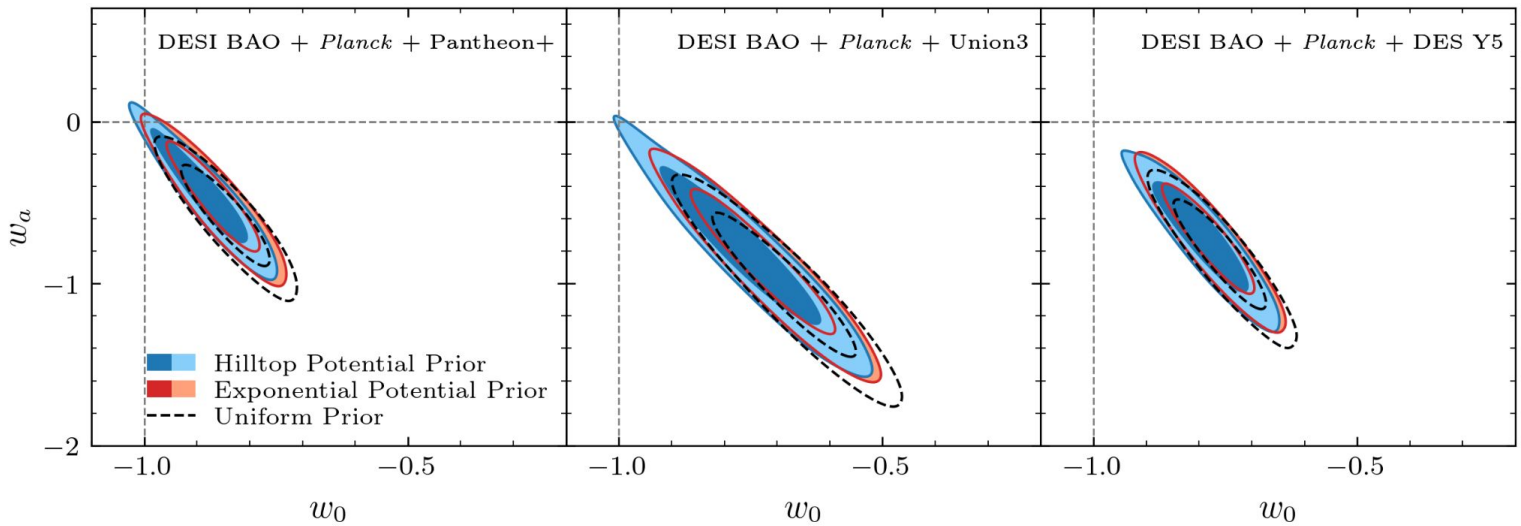
With theory-informed priors the preference for evolving DE decreases by **~2σ**



Toomey, **Montefalcone**, McDonough, Freese 2025

Theory-Informed Priors
Impact on the Preference for Evolving DE

When adding SNe data, the sensitivity to priors is significantly reduced to **≲1σ**



Toomey, **Montefalcone**, McDonough, Freese 2025